

How to handle LLMs in your research

Pablo Mosteiro, Robert Bagheri, Özgür Togay, Laurence Frank
Partly generated with help of Qwen/Qwen2.5-7B-Instruct and moonshotai/Kimi-K2-Instruct-0905 via together and groq, respectively (HuggingChat).

Department of Methodology, Statistics & Data Science

September 17, 2026

2026-09-17

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Today's Agenda

- Introduction (3 min)

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How to handle LLMs in your research

└ Today's Agenda

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- Introduction (3 min)
- How LLMs Work (10 min)



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- How LLMs Work (10 min)
- Limitations and Risks (10 min)



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- Prompting CAs and Chatbots (10 min)



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- Practical LLM Use in Research (10 min)

About us



<https://nlp.sites.uu.nl/>

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How to handle LLMs in your research

└ Intro

└ About us

About us

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Introduction: What do you know about LLMs?

- **Activity: Think-Pair-Share** (2 min)



Image produced in DALL-E December 2023 (Credit: Christine Fox)



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How to handle LLMs in your research

└ Intro

└ Introduction: What do you know about LLMs?

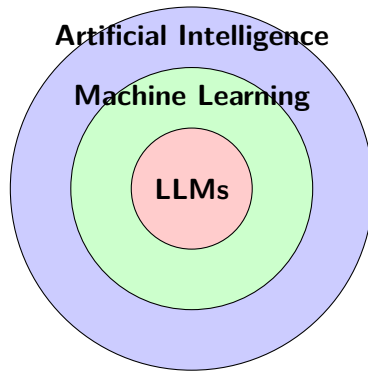
Introduction: What do you know about LLMs?

● Activity: **Think-Pair-Share** (2 min)



Image produced in DALL-E December 2023 (Credit: Christine Fox)

AI vs ML vs LLM: The Relationship



LLMs are a specific type of ML,
which is a subset of AI



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└ How LLMs work

└ AI vs ML vs LLM: The Relationship

Artificial Intelligence, ML, and LLMs: Key Definitions

Artificial Intelligence (AI)

Computer systems performing tasks requiring human intelligence

Examples: Chess playing, image recognition, language translation

Machine Learning (ML)

Subset of AI using algorithms that improve through experience

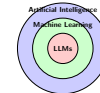
Learns patterns from data without explicit programming

Large Language Models (LLMs)

AI systems trained on vast text amounts

Understand and generate human language

Examples: GPT, Claude, Qwen, Mistral



LLMs are a specific type of ML,
which is a subset of AI

Model Learning Process: Training Phase

Training Phase

- Model learns patterns from large datasets

Example: LLM learns that "cat" often appears with "dog" in pet-related contexts



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└─How LLMs work

└─Model Learning Process: Training Phase

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Training Phase

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- └ How LLMs work
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- Requires massive computational resources (weeks / months)

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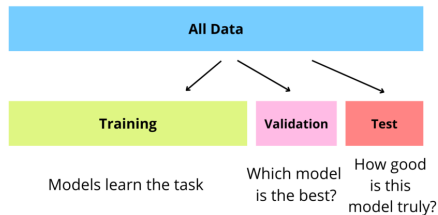
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Model Learning Process: Validation Phase

Validation and Testing



Source: <https://dqlab.id/5-panduan-praktis-membuat-model-machine-learning>

Key insight: Good models **generalize**, not just **memorize**



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How to handle LLMs in your research

└ How LLMs work

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During validation:

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Prompting phase

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Credit for this slide: Albert Gatt & Dong Nguyen

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- How LLMs work

└ Prompting phase

Prompting phase

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Credit for this slide: Albert Gatt & Dong Ngu

Prompting phase

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└ How LLMs work

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U weet



Credit for this slide: Albert Gatt & Dong Nguyen

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U weet wat



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How to handle LLMs in your research

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Credit for this slide: Albert Gatt & Dong Nguyen

What is Generative AI?

Definition: AI that creates new content (text, images, code, audio)

Key Features:

- Creates novel outputs, not just classifies

Examples: ChatGPT (text), DALL-E (images), CodeT5 (code)



How to handle LLMs in your research

└─ How LLMs work

└─ What is Generative AI?

LLMs are a form of GenAI (they are generative after all)

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Text-GenAI components

	consists of	fed by	GPT learns
	LLM	textual internet data	“knowledge” of language and content
	Dialogue Model	social media, forums	“knowledge” of dialogue structure
	Task Model	programmers	tasks, self knowledge and limits
	Ethical Governor	Kenian data labellers	ethical answers

Credit for this slide: Davitze Könning

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How to handle LLMs in your research

└─How LLMs work

└─Text-GenAI components

“Knowledge” b/c chatbots have no real knowledge or understanding; they produce what we want to hear based on predictions.

As a result, chatbots are better at being linguistically correct than at being content-wise correct.

The programmers also play a major role. You notice that adjustments are being made.

If you ask certain questions, the chatbots no longer want to answer them.

For example, there were many errors regarding the generation of incorrect sources.

LLM = Self-supervised learning

DM = learns how people interact with each other through social media, conversation data to make ChatGPT more human

TM = How to handle specific tasks, such as writing code, making plans, etc

EG = Filters, guidelines, and human supervision

Kenyan data labelers: come back to this point later

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Credit for this slide: Davitze Könning

Generative AI vs. Other Models

Feature	Generative	Discriminative
Primary Task	Create new content	Classify existing data
Output Type	Novel text/images	Labels/categories
Example	GPT, DALL-E	Spam filters, medical diagnosis



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How to handle LLMs in your research

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Activity: True or False?

Instructions: Raise your hand if you think the statement is TRUE

- **Statement 1:** Generative AI can only work with text



Activity: True or False?

Instructions: Raise your hand if you think the statement is TRUE

- **Statement 1:** Generative AI can only work with text
- **Statement 2:** All generative models are also discriminative



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Caveat

gold labels!



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- **Statement 3:** Generative AI creates completely original content never seen in training
- **Statement 4:** LLMs are a type of generative AI



Limitations and Risks: Hallucinations

What are hallucinations?

- LLMs generate false or nonsensical information (fake citations, non-existent facts)



How to handle LLMs in your research

└ Limitations and Risks

└ Limitations and Risks: Hallucinations

Example: Asking for “recent studies on X” might return made-up citations

Risk: Researchers might unknowingly use false information

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Limitations and Risks: Hallucinations

What are hallucinations?

- LLMs generate false or nonsensical information (fake citations, non-existent facts)
- Appears confident and plausible



Example: Asking for “recent studies on X” might return made-up citations

Risk: Researchers might unknowingly use false information



Bias

- **Gender bias:** Assumes doctors are male, nurses are female

Real-world impact: Biased models can perpetuate harmful stereotypes



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How to handle LLMs in your research

└─ Limitations and Risks

└─ Bias

Spend some time on each of the types of bias. Make sure they are clear.
Caused by input data or model architecture.

Bias

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Real-world impact: Biased models can perpetuate harmful stereotypes

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Limitations and Risks: Output Reliability

Reliability Issues

- **Inconsistency:** Same prompt can yield different answers

Give me the answer to the question "What is life?"

The question "What is life?" is a profound and complex one that has been explored by philosophers, scientists, theologians, and thinkers throughout history. There isn't a single, universally accepted answer, but here are some perspectives:

Give me the answer to the question "What is life?"

The question "What is life?" is a profound and complex one that has been explored by various disciplines, including biology, philosophy, theology, and more. Here are some key perspectives:

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How to handle LLMs in your research

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Limitations and Risks: Output Reliability

Reliability Issues

- **Inconsistency:** Same prompt can yield different answers
- **Context dependency:** Performance varies by domain

Pre-Chorus:

Tus pasos suenan como un susurro,
En la noche, bajo el manto oscuro,
Y en tus brazos, siento el fuego puro,
De un amor que nunca se apuro.

Best practice: Always verify critical information from multiple sources

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Limitations and Risks: Output Reliability

Reliability Issues

- **Inconsistency:** Same prompt can yield different answers
- **Context dependency:** Performance varies by domain
- **Prompt sensitivity:** Small changes affect outputs significantly

Write a Python program to do something basic

Certainly! Here is a simple Python program that calculates the sum of two numbers entered by the user:

```
# This is a simple Python program to calculate the sum of two numbers

def main():
    # Prompt the user to enter the first number
    num1 = float(input("Enter the first number: "))

    # Prompt the user to enter the second number
    num2 = float(input("Enter the second number: "))

    # Calculate the sum of the two numbers
    sum_result = num1 + num2

    # Print the result
    print(f"The sum of {num1} and {num2} is {sum_result}")

if __name__ == "__main__":
    main()
```

Write a Python program to do something very basic

Certainly! Here is a simple Python program that prints "Hello, World!" to the console. This is a classic example often used to introduce programming beginners to the basics of a programming language:

```
# This is a simple Python program to print "Hello, World!"

def main():
    # Print "Hello, World!" to the console
    print("Hello, World!")

if __name__ == "__main__":
    main()
```

Best practice: Always verify critical information from multiple sources

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Limitations and Risks: Ethics

Ethical Concerns

- **Data:** Training data may include personal or copyrighted material

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How to handle LLMs in your research

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Limitations and Risks: Ethics

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How to handle LLMs in your research

└─ Limitations and Risks

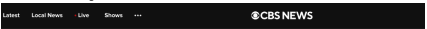
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Limitations and Risks: Ethics

Ethical Concerns

- **Data:** Training data may include personal or copyrighted material
- **Environmental impact:** High energy consumption
- **Labor practices:** Data annotation often involves poor working conditions



"When it gets to the day before payday, they close the account and say that you violated a policy," Kanyugi said.

Employees say they have no recourse or even a way to complain.

The company told 60 Minutes that any work done "in line with our community guidelines was paid out." In March, as workers started complaining publicly, Remotasks abruptly shut down in Kenya, locking all workers out of their accounts.

The mental toll of AI training

Workers say some of the projects for Meta and OpenAI also caused them mental harm. Wambalo was assigned to train AI to recognize and weed out pornography, hate speech and excessive violence from social media. He had to sift through the worst of the worst content online for hours on end.

"I looked at people being slaughtered," Wambalo said. "People engaging in sexual activity with animals. People abusing children physically, sexually. People committing suicide."

Berhane Gebrekidan thought she'd been hired for a translation job, but she said what she ended up doing was reviewing content featuring dismembered bodies and drone attack victims.

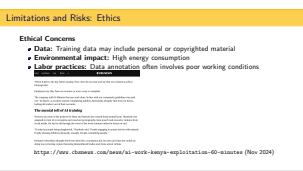
<https://www.cbsnews.com/news/ai-work-kenya-exploitation-60-minutes> (Nov 2024)

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How to handle LLMs in your research

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Limitations and Risks: Ethics

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Results

Cultural modulation of neural mechanisms asserts that technological interactions significantly influence brain development and cognitive abilities. The use of artificial intelligence chatbots like ChatGPT for cognitive offloading may lead to underemployment of specific cognitive faculties, inhibiting their full maturation. This phenomenon is particularly relevant in the context of executive functions, where reliance on artificial intelligence for problem-solving can reduce cognitive effort and lead to long-term cognitive changes.¹³

- **Effect on cognition:**

Dubey et al. Redefining Cognitive Domains in the Era of ChatGPT: A Comprehensive Analysis of Artificial Intelligence’s Influence and Future Implications. Med Res Arch. 2024 Jun;12(6):5383

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Effective Prompting: BRAVE(R) Framework

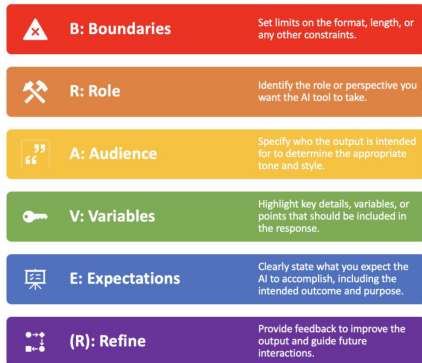
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└ Prompting

└ Effective Prompting: BRAVE(R) Framework

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Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).

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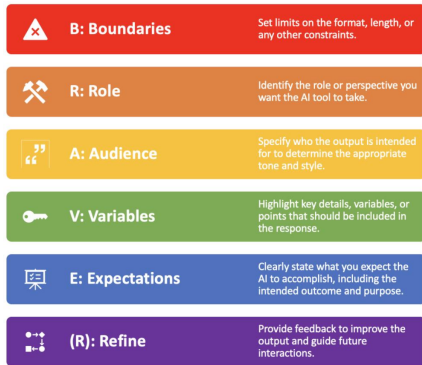
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Effective Prompting: BRAVE(R) Framework



Effective Prompting: BRAVE(R) Framework



Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).

Poor prompt: “Summarize this paper”

Better prompt: “Act as a psychology research assistant [R]. Summarize this paper in 3 bullet points [B] focusing on methodology and findings relevant to cognitive behavioral therapy [V]. Use academic language [A] and APA style [E].”

Key insight: Specific, contextual prompts yield better results

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How to handle LLMs in your research

- └ Prompting

- └ Effective Prompting: BRAVE(R) Framework

Effective Prompting: BRAVE(R) Framework



Poor prompt: “Summarize this paper”
Better prompt: “Act as a psychology research assistant [R]. Summarize this paper in 3 bullet points [B] focusing on methodology and findings relevant to cognitive behavioral therapy [V]. Use academic language [A] and APA style [E].”
Key insight: Specific, contextual prompts yield better results

Evaluating and Critically Assessing GenAI Responses

Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).



F: Focus

Focus on the AI's response by breaking it down into its main points and arguments.



A: Authenticate

Authenticate the AI's output by verifying it against current research and empirical data.



C: Critique

Critique the response's accuracy, relevance, and depth in the context of the topic.



T: Think

Think about the response from various perspectives and its potential impact.



S: Scrutinise

Scrutinise the AI's conclusions by critically questioning and exploring alternative hypotheses or interpretations.

More info:

<https://teachersguides1.nl/genai/generative-ai-guidelines/>



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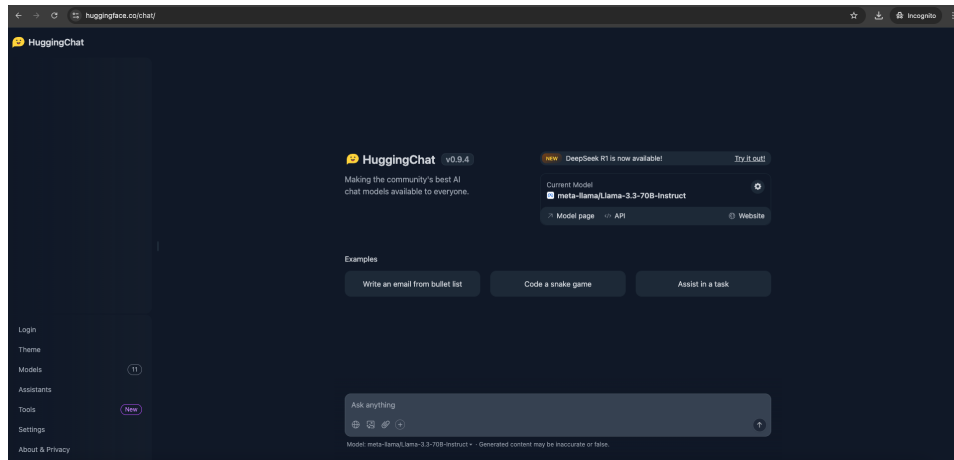
How to handle LLMs in your research

└ Prompting

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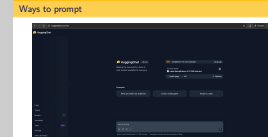
Ways to prompt



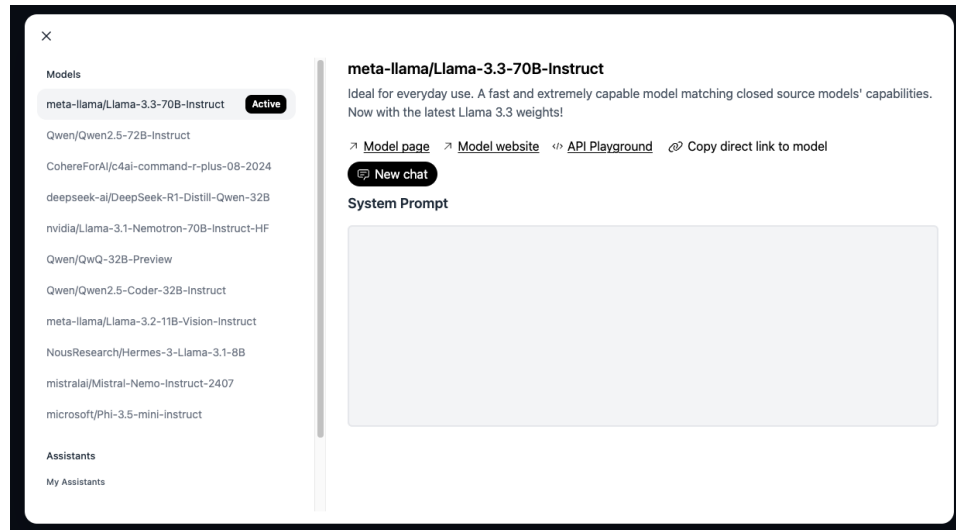
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- └ Prompting
 - └ Ways to prompt



Switch models



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How to handle LLMs in your research

└ Prompting

└ Switch models

Using an LLM locally (no server)

But model can still be train in a very closed-source way



Official UU policy

`https://www.uu.nl/en/organisation/ai-policy/researchers`

- Benefits and use cases



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└ Practical LLM Use in Research

└ Official UU policy

I don't want to go into details because this changes all the time.

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- Benefits and use cases
- Risks and limitations



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- Benefits and use cases
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- Effective use of GenAI (*Background check*)



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- Benefits and use cases
- Risks and limitations
- Effective use of GenAI (*Background check*)
- Citing GenAI in your work



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How to handle LLMs in your research

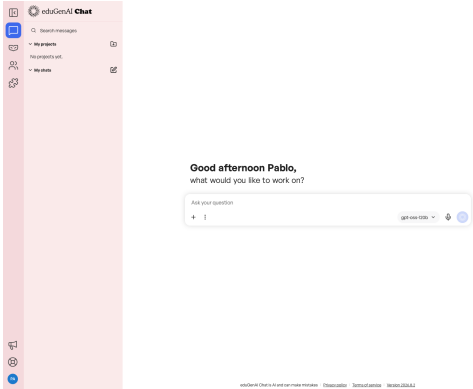
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EduGenAI

`https://www.uu.nl/en/organisation/ai-policy/teachers/edugenai`



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How to handle LLMs in your research

└ Practical LLM Use in Research

└ EduGenAI



UU AI-chat

<https://www.uu.nl/en/organisation/ai-policy/ai-platforms-at-uu>

UU AI-chat

UU AI-Chat is Utrecht University's own generative AI chat environment. You use it via your browser to have conversations with various language models, within an environment managed by the University itself. The platform is based on the University of Amsterdam's AI chat environment. UU AI-Chat will be available in early September.

Who is it for?

It is intended for all students and staff at Utrecht University, for use in teaching, research and administrative operations. Guests and external parties cannot use the service.

What can you do with it?



More about UU AI Chat



How do you gain access?

This service will be available to students and staff from the start of September.



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LLM Use in Research

Appropriate & Inappropriate Uses

- Think-Pair-Share



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How to handle LLMs in your research

└ Practical LLM Use in Research

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LLM Use in Research: Appropriate Applications

Appropriate Uses

- **Literature review:** Summarizing papers (with verification)

Key: Always maintain human oversight and verification



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LLM Use in Research: Appropriate Applications

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Caveat

only when you know which statistical analysis is suitable

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LLM Use in Research: Inappropriate Applications

Avoid Using LLMs For:

- **Statistical analysis:** LLMs can't perform proper statistical tests



LLM Use in Research: Inappropriate Applications

Avoid Using LLMs For:

- **Statistical analysis:** LLMs can't perform proper statistical tests
- **Fact-checking:** They may hallucinate information



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What can go wrong?
Encourage student assistants, PhD candidates and postdocs to triple-check outputs of GenAI. GenAI's coding skills are great, but if we don't scrutinize what it does, we don't understand the intermediate steps, and it becomes harder to make choices. Especially when choosing methodologies.

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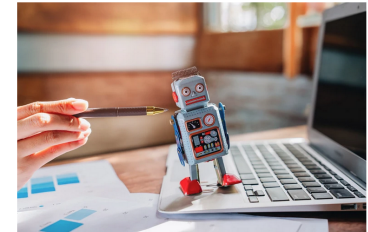
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Don't let your students use AI as a ghostwriter

How a research proposal generated by artificial intelligence transformed my approach to teaching and supervision.

By Yanjun Shen





Using generative artificial intelligence models to kick ideas around is more effective than outsourcing thinking wholesale, says Yanjun Shen. Credit: iahkitti-hai/Stock via Getty

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A thumbnail image of the article 'Don't let your students use AI as a ghostwriter' from Nature. It shows a small robot on a laptop keyboard, similar to the main image.

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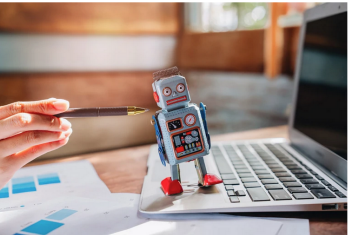
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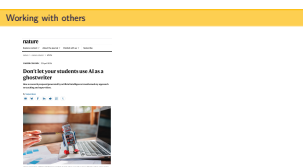




Using generative artificial intelligence models to kick ideas around is more effective than outsourcing thinking wholesale, says Yanjun Shen. Credit: iahkittihaiStock via Getty

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A hand holding a pen points towards a small, blue and red robot standing on a laptop keyboard. The robot has a boxy head with large eyes and a small antenna. The background is a blurred desk with a laptop and some papers.

Using generative artificial intelligence models to kick ideas around is more effective than outsourcing thinking wholesale, says Yanjun Shen. Credit: iahkittihai/Stock via Getty

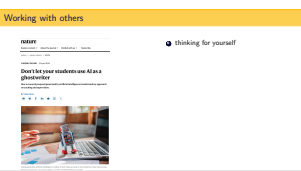
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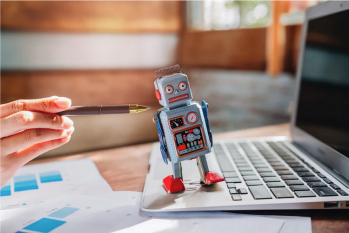
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- ① thinking for yourself
- ② read literature

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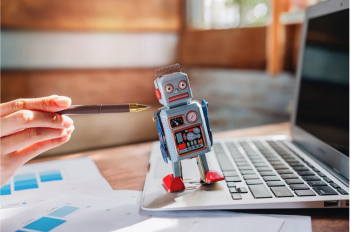
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- 1 thinking for yourself
- 2 read literature
- 3 ask AI for advice

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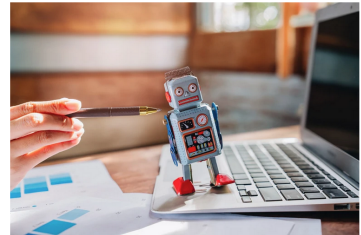
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